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The equation $2x^4 + xy + x + y^3 - y + 1 = 0$ has no integer solutions

Size $h = 49$, length $l = 12$

Theorem 1. *The equation*

$$2x^4 + xy + x + y^3 - y + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses the cubic Hilbert reciprocity law over the field $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. It uses 2 graphs. For the g -th graph, we attach to a hypothetical integer solution the values of auxiliary polynomials $H_i^{(g)}$ with coefficients in $\mathbb{Z}[\omega]$, constants $c_i^{(g)} \in \mathbb{Z}[\omega]$, and a sum $B^{(g)}$ of cubic Hilbert symbols of these values, recorded by weighted edges. By the reciprocity law, the local contributions $B_p^{(g)}$ of all primes p add up to 0 modulo 3, for every g . We show that $B_p^{(g)} = 0$ for $p \notin S$, compute the vectors $(B_p^{(g)})_g$ for the primes $p \in S$, and obtain a contradiction. Two standard facts from class field theory are quoted without proof: the explicit formula for the cubic Hilbert symbol at the places not above 3, and the Hilbert reciprocity law. Apart from these, the proof uses polynomial identities, which can be checked by expanding both sides, and finite computations with residues.

1 Cubic Hilbert symbols over $\mathbb{Q}(\omega)$

The field. Let $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. Its ring of integers is $\mathbb{Z}[\omega]$, and every element of K can be written uniquely as $A + B\omega$ with $A, B \in \mathbb{Q}$. We write

$$\langle A, B \rangle = A + B\omega,$$

also when A and B are polynomials in x and y . Then $\langle A, B \rangle \langle C, D \rangle = \langle AC - BD, AD + BC - BD \rangle$, and the norm is $N(\langle A, B \rangle) = A^2 - AB + B^2$. In particular, a value $\langle A, B \rangle$ with $A, B \in \mathbb{Z}$ is zero if and only if $A = B = 0$.

Places. The finite places of K correspond to the non-zero prime ideals of $\mathbb{Z}[\omega]$. A prime $p \equiv 2 \pmod{3}$ is inert: there is one place above p , with uniformizer p and residue field $\mathbb{Z}[\omega]/p\mathbb{Z}[\omega]$ with p^2 elements. If $p \equiv 1 \pmod{3}$, the polynomial $t^2 + t + 1$ has two roots r modulo p , and for each of them there is a place v_r above p , at which ω is the root of $t^2 + t + 1$ in $\mathbb{Z}_p$ congruent to r modulo p ; the completion is $\mathbb{Q}_p$, the uniformizer is p , the residue field is $\mathbb{F}_p$, and ω reduces to r . The prime 3 is ramified: $\pi = 1 - \omega$ generates the unique prime ideal above 3, $\pi^2 = -3\omega$, and $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega] = \mathbb{F}_3$, where $\langle A, B \rangle$ reduces to $A + B$. The only infinite place of K is complex. For a finite place v we write K_v for the completion and $v(a)$ for the normalized valuation, and we call a a *unit* at v if $v(a) = 0$.

Hilbert symbols. For a place v of K and $a, b \in K_v^\times$, the cubic Hilbert symbol $(a, b)_v$ is a cube root of unity, see [1, Chapter V, §3]. We write $(a, b)_v = \omega^{\beta_v(a, b)}$ with $\beta_v(a, b) \in \mathbb{Z}/3\mathbb{Z}$, and we use the following three facts [1, Chapter V, §3 and Chapter VI, §8].

(H1) $\beta_v(a, bc) = \beta_v(a, b) + \beta_v(a, c)$, $\beta_v(a, b) = -\beta_v(b, a)$, and $\beta_v(a, c^3) = 0$. In particular, $\beta_v(a, b)$ depends only on the classes of a and b in $K_v^\times/K_v^{\times 3}$.

(H2) Let v be a finite place not above 3, let q be the number of elements of its residue field, and let ϖ be a uniformizer at v . Write $a = \varpi^e u$ and $b = \varpi^f w$ with units u and w , and define $\chi(u) \in \mathbb{Z}/3\mathbb{Z}$ by $\bar{u}^{(q-1)/3} = \bar{\omega}^{\chi(u)}$, where the bar denotes the reduction to the residue field. Then $\beta_v(a, b) = e\chi(w) - f\chi(u)$. In particular, $\beta_v(a, b) = 0$ if a and b are both units.

(H3) (Hilbert reciprocity.) For $a, b \in K^\times$, $\beta_v(a, b) = 0$ for all but finitely many places v , and $\sum_v \beta_v(a, b) = 0$, where the sum is over all places of K . At the complex place, $\beta_v(a, b) = 0$.

The general formula in (H2) contains an additional term $ef\chi(-1)$, which vanishes because $-1 = (-1)^3$ is a cube. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values β_v change sign, and the proof below is the same.

Classes at the places not above 3. Let $p \neq 3$ be a prime and v a place above p . For $a = p^e u \in K_v^\times$ with a unit u , put $c_v(a) = (e \bmod 3, \chi(u)) \in \mathbb{F}_3^2$. By (H2) and (H1), c_v is additive, $c_v(a)$ determines the class of a modulo cubes, and $\beta_v(a, b) = e\chi(w) - f\chi(u)$ for $c_v(a) = (e, \chi(u))$ and $c_v(b) = (f, \chi(w))$. We put $c_p(a) = c_v(a) \in \mathbb{F}_3^2$ if $p \equiv 2 \pmod{3}$, and $c_p(a) = (c_{v_r}(a), c_{v_{r'}}(a)) \in \mathbb{F}_3^4$ if $p \equiv 1 \pmod{3}$, where $r < r'$ are the roots of $t^2 + t + 1$ in $\{1, \dots, p-1\}$; accordingly, β_p is the sum of the forms β_v over the places v above p . For $p \equiv 2 \pmod{3}$ and a rational integer u prime to p , we have $\bar{u} \in \mathbb{F}_p$, so that $\bar{u}^{(p^2-1)/3} = (\bar{u}^{p-1})^{(p+1)/3} = 1$ and $\chi(u) = 0$.

The place above 3.

Lemma 2. *Let v be the place above 3. Every non-zero element of K_v is equal to $\pi^a \omega^b 2^c (1 + 3\omega)^d$ times a cube, for unique $a, b, c, d \in \mathbb{Z}/3\mathbb{Z}$, and every unit congruent to 1 modulo 9 is a cube. Put $c_3(A) = (a, b, c, d) \in \mathbb{F}_3^4$. Then c_3 is additive, and $\beta_v(A, B) = c_3(A)^\top M_3 c_3(B)$, where*

$$M_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \pmod{3}.$$

Proof. For $\alpha \in \mathbb{Z}[\omega]$, the polynomial $g(s) = s + 3s^2 + 3s^3 - \alpha$ satisfies $g'(s) \equiv 1 \pmod{\pi}$, so by Hensel's lemma it has a root s in the completion of $\mathbb{Z}[\omega]$, and then $(1 + 3s)^3 = 1 + 9(s + 3s^2 + 3s^3) = 1 + 9\alpha$. Hence every unit congruent to 1 modulo 9 is a cube in K_v , and the class of a unit modulo cubes depends only on its residue modulo 9. There are 54 units in $\mathbb{Z}[\omega]/(9)$, and a direct enumeration shows that the 27 products $\omega^b 2^c (1 + 3\omega)^d$ with $b, c, d \in \{0, 1, 2\}$ are pairwise distinct modulo 9, also after multiplication by $-1 = (-1)^3$. Hence they represent all 27 classes of units modulo cubes. Since π^3 is a cube times a unit, the exponent of π is the valuation modulo 3, and the first statement follows.

By (H1), it remains to compute β_v on the 16 ordered pairs of the elements $\pi, \omega, 2$ and $1 + 3\omega$. Their norms are 3, 1, 4 and 7, hence, for each such pair, all the symbols at the places not above 2, 3 and 7 vanish by (H2), and, by (H3), β_v at the place above 3 is minus the sum of the values at the place above 2 and at the two places above 7. These places correspond to $\bar{\omega} = 2$ and $\bar{\omega} = 4$ in $\mathbb{F}_7$. The classes (e, χ) of the four elements at these places are

	2	$\bar{\omega} = 2$	$\bar{\omega} = 4$
π	(0, 2)	(0, 0)	(0, 2)
ω	(0, 1)	(0, 2)	(0, 2)
2	(1, 0)	(0, 2)	(0, 1)
$1 + 3\omega$	(0, 2)	(1, 0)	(0, 0)

where, at the places above 7, $\chi(u)$ is defined by $\bar{u}^2 = \bar{\omega}^{\chi(u)}$ in $\mathbb{F}_7$, and 7 is a uniformizer. Formula (H2) and the sum over these three places give M_3 . $\square$

We write β_3 for the form of Lemma 2. Since $3 = -\omega^2\pi^2$ and $4 = 2^2$, we have $c_3(3) = (2, 2, 0, 0)$ and $c_3(4) = (0, 0, 2, 0)$.

Residue classes. The local computations use the following description of the classes of the values of a polynomial on a residue class.

Lemma 3. *Let p be a prime, $k \geq 1$, and let $z \in \mathbb{Z}[\omega]$ be non-zero with $z \equiv \zeta \pmod{p^k}$, where $\zeta = \langle A, B \rangle$ with $0 \leq A, B < p^k$. Then $c_p(z) \in b + D$ for the vector b and the subspace D given as follows.*

- (i) $p \equiv 2 \pmod{3}$. If $\zeta \neq 0$, then $b = c_p(\zeta)$ and $D = 0$. If $\zeta = 0$, then $b = 0$, and D is $\mathbb{F}_3^2$, or the span of $(1, 0)$ if $z \in \mathbb{Z}$.
- (ii) $p \equiv 1 \pmod{3}$. For each root r , let ρ be the root of $t^2 + t + 1$ modulo p^k congruent to r . If $A + B\rho \not\equiv 0 \pmod{p^k}$, the two coordinates of $c_{v_r}(z)$ are those of $c_{v_r}(\zeta)$; otherwise these two coordinates are arbitrary.
- (iii) $p = 3$, $z \notin \mathbb{Z}$. If $\zeta \neq 0$, let $a = v(\zeta) < 2k$ be its valuation at π and $m = 2k - a$. Then $b = c_3(\zeta)$, and $D = 0$ if $m \geq 4$, while $D = D_m$ if $m \leq 3$, where D_1 is spanned by $(0, 1, 0, 0)$, $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, D_2 by $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, and D_3 by $(0, 0, 1, 1)$. If $\zeta = 0$, then $b = 0$ and $D = \mathbb{F}_3^4$.
- (iv) $p = 3$, $z \in \mathbb{Z}$, so that $B = 0$. If $3^k \nmid A$, let $j = v_3(A)$. Then $b = c_3(A)$ and $D = 0$ if $k - j \geq 2$, while $b = j c_3(3)$ and D is spanned by $c_3(4)$ if $k - j = 1$. If $3^k \mid A$, then $b = 0$ and D is spanned by $c_3(3)$ and $c_3(4)$.

Proof. (i) If $\zeta \neq 0$, then $e = v(z) = v(\zeta) < k$ and $z/p^e \equiv \zeta/p^e \pmod{p}$, so that χ is determined. If $z \in \mathbb{Z}$, its unit part is a rational integer, and $\chi = 0$. (ii) At v_r , z and ζ are the p -adic numbers $A' + B'\omega$ and $A + B\omega$ with $\omega \equiv \rho \pmod{p^k}$, and the argument of (i) applies. (iii) If $\zeta \neq 0$, then $v(z) = a$, and $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$. By Lemma 2, the class of a unit depends only on its residue modulo $9 = \omega\pi^4$; the units congruent to 1 modulo π^m form a subgroup, and a direct enumeration of their residues modulo 9 shows that their classes form D_m . (iv) If $3^k \nmid A$, then $z = 3^j u$ with a rational unit $u \equiv A/3^j \pmod{3^{k-j}}$. The class of a rational unit depends only on its residue modulo 9, the units ± 1 are cubes, and the rational units congruent to 1 modulo 3 are 4^s times a unit congruent to 1 modulo 9; hence the classes are as stated, and if $3^k \mid A$ the valuation j is also unknown. $\square$

Lemma 4. *Let p be a prime, and consider vertices i with vectors b_i , subspaces D_i and vectors γ_i , and a set of edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2\}$. Put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij}$ for the edges, and $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge (i, j) and all $d \in D_i$, $d' \in D_j$. Then*

$$\sum_{(i,j)} w_{ij} \beta_p(c_i, c_j) + \sum_i \beta_p(c_i, \gamma_i) = \sum_{(i,j)} w_{ij} \beta_p(b_i, b_j) + \sum_i \beta_p(b_i, \gamma_i)$$

whenever $c_i \in b_i + D_i$ for every i . By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and $\beta_p(u, v) = -\beta_p(v, u)$, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{(i,j)} w_{ij} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

2 The graphs

Put $F(x, y) = 2x^4 + xy + x + y^3 - y + 1$, so that (1) is the equation $F(x, y) = 0$. There are 2 certificates. The g -th certificate consists of polynomials $H_i^{(g)}$ with coefficients in $\mathbb{Z}[\omega]$ and constants $c_i^{(g)} \in \mathbb{Z}[\omega]$, listed in Table 1 and Table 2, and of edges (i, j) with $i < j$ and weights $w_{ij}^{(g)} \in \{1, 2\}$. For an edge we put $\rho_{ij}^{(g)} = w_{ij}^{(g)}$ and $\rho_{ji}^{(g)} = -w_{ij}^{(g)} \bmod 3$, and $\rho_{ij}^{(g)} = 0$ otherwise. In the rest of the proof we usually suppress the index g . The edges $(i, j; w_{ij})$ of the first certificate are (0, 1; 2), (0, 4; 1), (0, 5; 1), (0, 7; 1), (0, 10; 1), (0, 13; 2), (0, 15; 1), (0, 16; 2), (0, 18; 1), (0, 27; 2), (1, 2; 2), (1, 3; 2), (1, 5; 1), (1, 8; 1), (1, 11; 1), (1, 12; 2), (1, 15; 1), (1, 23; 1), (2, 5; 2), (2, 9; 2), (2, 16; 2), (2, 17; 2), (3, 12; 2), (3, 16; 1), (3, 18; 2), (3, 19; 1), (3, 23; 1), (4, 18; 2), (4, 23; 1), (4, 24; 1), (4, 27; 2), (5, 9; 1), (5, 19; 1), (5, 23; 2), (6, 9; 2), (6, 10; 1), (6, 11; 2), (6, 18; 1), (6, 21; 1), (6, 28; 2), (7, 9; 2), (7, 14; 1), (7, 19; 1), (8, 9; 1), (8, 15; 2), (8, 16; 1), (9, 14; 2), (9, 20; 2), (9, 22; 1), (10, 20; 1), (10, 22; 1), (10, 25; 2), (11, 15; 2), (11, 16; 1), (11, 17; 2), (11, 22; 1), (11, 27; 2), (12, 13; 1), (12, 14; 1), (12, 28; 2), (13, 14; 1), (13, 15; 2), (13, 16; 2), (13, 21; 2), (13, 27; 2), (14, 15; 1), (15, 19; 1), (17, 21; 2), (17, 27; 1), (18, 24; 2), (18, 26; 1), (19, 24; 1), (19, 26; 2), (20, 21; 1), (21, 25; 1), (22, 28; 2), (23, 26; 1), (24, 26; 1), (24, 27; 1), (26, 28; 2). The edges $(i, j; w_{ij})$ of the second certificate are (0, 15; 1), (0, 26; 1), (1, 6; 2), (1, 11; 2), (1, 12; 1), (1, 22; 2), (2, 5; 2), (2, 6; 1), (2, 7; 2), (2, 8; 1), (2, 10; 1), (2, 11; 1), (2, 13; 1), (2, 17; 1), (2, 18; 1), (2, 19; 2), (2, 21; 1), (2, 26; 1), (2, 29; 2), (3, 4; 2), (3, 8; 1), (3, 16; 1), (3, 21; 2), (3, 23; 1), (3, 25; 2), (4, 5; 1), (4, 17; 1), (4, 20; 1), (4, 21; 1), (4, 22; 1), (4, 23; 1), (4, 29; 2), (5, 10; 1), (5, 14; 2), (5, 16; 1), (5, 18; 1), (5, 19; 1), (5, 23; 2), (5, 24; 1), (5, 25; 1), (5, 31; 1), (6, 12; 1), (6, 17; 2), (6, 19; 1), (6, 20; 1), (6, 27; 2), (6, 30; 2), (7, 8; 1), (7, 9; 2), (7, 14; 1), (7, 16; 2), (7, 17; 1), (7, 19; 1), (7, 22; 2), (7, 25; 1), (7, 29; 1), (8, 9; 1), (8, 13; 1), (8, 15; 2), (8, 17; 2), (8, 19; 2), (8, 21; 2), (8, 22; 2), (8, 29; 1), (9, 21; 1), (10, 15; 2), (10, 27; 1), (10, 30; 2), (11, 18; 1), (11, 20; 2), (11, 21; 2), (11, 24; 2), (11, 27; 1), (12, 23; 1), (12, 25; 2), (12, 26; 2), (12, 30; 2), (13, 18; 2), (13, 21; 2), (13, 22; 2), (13, 25; 1), (13, 26; 2), (14, 15; 2), (14, 21; 2), (14, 22; 1), (14, 26; 1), (15, 22; 2), (15, 27; 1), (16, 21; 2), (16, 25; 1), (16, 26; 1), (16, 27; 2), (16, 28; 2), (16, 29; 2), (17, 19; 2), (17, 20; 2), (17, 21; 1), (17, 22; 2), (17, 30; 2), (18, 19; 2), (18, 23; 2), (18, 24; 1), (18, 26; 1), (18, 30; 2), (19, 21; 2), (20, 21; 2), (21, 26; 2), (21, 31; 1), (22, 28; 2), (23, 26; 2), (23, 28; 2), (24, 28; 1), (24, 31; 2), (25, 30; 2), (26, 28; 1), (26, 30; 1), (26, 31; 2), (27, 28; 1). Let (ξ, η) denote (x, y) or (y, x) . We call a vertex i linear if $H_i = \varepsilon_i(\eta - r_i(\xi))$ for a unit ε_i of $\mathbb{Z}[\omega]$ and a polynomial r_i with coefficients in $\mathbb{Z}[\omega]$; the last column of Table 1 and its analogues gives $\eta = r_i(\xi)$ for the linear vertices. For a linear vertex i , we put $f_i(t) = F$ and $g_{ij}(t) = H_j$ evaluated at $\xi = t$, $\eta = r_i(t)$; these are polynomials in t with coefficients in $\mathbb{Z}[\omega]$. For every linear vertex i , Table 3 gives an integer $D_i \geq 1$ and a polynomial $w_i(t)$ with coefficients in $\mathbb{Z}[\omega]$ such that

$$D_i^3 c_i \prod_j g_{ij}^{\rho_{ij}} - w_i^3 = f_i Q_i \quad (2)$$

for a polynomial $Q_i(t)$ with coefficients in $\mathbb{Z}[\omega]$, where $\rho_{ij} \in \{0, 1, 2\}$; Q_i is the quotient of the division of the left side by f_i , and the remainder is zero. For every edge (i, j) with a linear endpoint, Table 4 gives the first linear endpoint $k \in \{i, j\}$ and the factorization of the norm of the resultant $R_{ij} = \text{Res}_t(f_k, g_{kl})$, where $\{k, l\} = \{i, j\}$; it is non-zero. These identities can be checked by expanding both sides and by polynomial division, using $\omega^2 = -1 - \omega$. They hold for each certificate, with its own data.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) , which lie in $\mathbb{Z}[\omega]$.

Step 1: the values are non-zero. The vertex H_{11} of the first certificate is linear, and $H_{11} = 0$ would give $f_{11}(\xi) = 0$ with $\xi \in \mathbb{Z}$, where $f_{11}(t) = \langle 2t^4 + t^3 + 4t^2 + 4t + 1, 0 \rangle$. An integer root of $2t^4 + t^3 + 4t^2 + 4t + 1$ divides its constant term 1, and $f_{11}(c) \neq 0$ for $c \in \{-1, 1\}$; hence $H_{11} \neq 0$. For $i \in \{0, 1, 2, 3, 4, 5, 7, 8, 9, 12, 13, 14, 15, 16, 17, 19, 20, 21, 23, 24, 25, 26, 27\}$ in the first certificate, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $H_i \neq 0$. For $i \in \{10, 28\}$ in the first certificate, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $H_i \neq 0$. For $i \in \{18, 22\}$ in the first certificate, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $H_i \neq 0$. For $i \in \{6\}$ in the first certificate, there is no residue pair (a, b) modulo 7 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{7}$, as one checks for the 7^2 residue pairs; hence $H_i \neq 0$. For $i \in \{1, 2, 5, 6, 7, 9, 10, 12, 14, 15, 16, 17, 20, 22, 23, 24, 27, 28, 29, 30\}$ in the second certificate, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $H_i \neq 0$. For $i \in \{0, 8, 18, 19, 21, 25, 31\}$ in the second certificate, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $H_i \neq 0$. For $i \in \{3, 4, 11\}$ in the second certificate, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $H_i \neq 0$. For $i \in \{26\}$ in the second certificate, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $H_i \neq 0$. For $i \in \{13\}$ in the second certificate, there is no residue pair (a, b) modulo 7 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{7}$, as one checks for the 7^2 residue pairs; hence $H_i \neq 0$. Also, all c_i are non-zero.

Step 2: reciprocity. For every place v of K , put

$$B_v^{(g)} = \sum_{(i,j)} w_{ij} \beta_v(H_i, H_j) + \sum_i \beta_v(H_i, c_i) \in \mathbb{Z}/3\mathbb{Z}, \quad (3)$$

the first sum over the edges of the g -th certificate, and, for a rational prime p , put $B_p^{(g)} = \sum_{v|p} B_v^{(g)}$. Applying (H3) to every pair (H_i, H_j) with an edge, multiplied by w_{ij} , and to every pair (H_i, c_i) , and adding, we find that $B_v^{(g)} = 0$ for all but finitely many v and $\sum_p B_p^{(g)} = 0$ for every g , as the complex place contributes 0. By (H1) and the definition of c_p , $B_p^{(g)} = \sum_{(i,j)} w_{ij} \beta_p(c_p(H_i), c_p(H_j)) + \sum_i \beta_p(c_p(H_i), c_p(c_i))$, where β_p is the form of Lemma 2 if $p = 3$.

Step 3: the primes outside S . Let $S = \{2, 3, 5, 7, 13\}$. It contains 3 and the primes dividing the norms of the constants c_i , of the D_i , of the R_{ij} . Let $p \notin S$, and let v be a place above p , with residue field k_v ; then all these numbers are units at v . First, no edge (i, j) has both values divisible by v . Indeed, suppose that v divides H_i and H_j . If R_{ij} is defined, let k be the linear endpoint used for it and $\{k, l\} = \{i, j\}$, and put $t = \xi$. As $H_k = \varepsilon_k(\eta - r_k(\xi))$, we have $\eta \equiv r_k(t)$ modulo v , hence $f_k(t) \equiv F(x, y) = 0$ and $g_{kl}(t) \equiv H_l \equiv 0$ modulo v . The resultant can be written as $R_{ij} = Af_k + Cg_{kl}$ with polynomials A and C with coefficients in $\mathbb{Z}[\omega]$, so v divides R_{ij} , a contradiction. By (H2), a term of (3) in which all arguments are units vanishes. Now let v divide H_i . Collecting the terms of (3) which contain H_i , and using (H1) and $\rho_{ji} = -w_{ij}$ for the edges (j, i) with $j < i$, we obtain $\beta_v(H_i, \Pi_i)$ with $\Pi_i = c_i \prod_j H_j^{\rho_{ij}}$, which is a unit. We claim that $D_i^3 \Pi_i \equiv W^3$ modulo v for some $W \in \mathbb{Z}[\omega]$. If i is linear, put $t = \xi$; then $g_{ij}(t) \equiv H_j$ and $f_i(t) \equiv 0$ modulo v as above, and (2) gives the claim with $W = w_i(t)$. Hence the reduction of Π_i is a non-zero cube in k_v , $\chi(\Pi_i) = 0$, and $\beta_v(H_i, \Pi_i) = v(H_i)\chi(\Pi_i) = 0$ by (H2). Hence $B_v^{(g)} = 0$, and $B_p^{(g)} = 0$.

Step 4: the primes in S . Let $p \in S$. Here 2 is inert, 5 is inert, the places above 7 are v_2 and v_4 , and the places above 13 are v_3 and v_9 . Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then

$H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 3, applied with ζ the residue of $H_i(a, b)$, gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; when H_i has rational coefficients, $H_i \in \mathbb{Z}$, and we use this in (i) and (iv). With $\gamma_i = c_p(c_i)$, if the hypotheses of Lemma 4 hold, then $B_p^{(g)}$ is determined by a , b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a$, $y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses hold for every certificate, or until no solution remains. Table 5 lists the resulting terminal classes and the value of the vector $(B_p^{(g)})_g$ in each; every integer solution lies in one of them. The result is $(B_2^{(g)})_g = (1, 2)$, $(B_3^{(g)})_g \in \{(0, 1), (0, 2), (1, 1), (1, 2), (2, 0)\}$, $(B_5^{(g)})_g = (0, 0)$, $(B_7^{(g)})_g = (0, 0)$, $(B_{13}^{(g)})_g = (0, 0)$.

Step 5: contradiction. By Steps 2 and 3, $\sum_{p \in S} B_p^{(g)} = 0$ for every g . But by Step 4, the vector $(\sum_{p \in S} B_p^{(g)})_g$ lies in $\{(0, 2), (1, 0), (1, 1), (2, 0), (2, 1)\}$, which does not contain $(0, 0)$. This contradiction proves the theorem. $\square$

References

- [1] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften 322, Springer, Berlin, 1999.

Table 1: The polynomials H_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$ of the first certificate. Here $\langle A, B \rangle = A + B\omega$.

i	H_i	c_i	chart
0	$\langle x^2 - y, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2, 0 \rangle$
1	$\langle x - y + 1, -x^2 + x + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x + 1, -x^2 + x + 1 \rangle$
2	$\langle -x^2 - y + 1, -x^2 + x + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 + 1, -x^2 + x + 1 \rangle$
3	$\langle x, 0 \rangle$	$\langle 0, -1 \rangle$	$x = \langle 0, 0 \rangle$
4	$\langle x + 1, 1 \rangle$	$\langle 0, -1 \rangle$	$x = \langle -1, -1 \rangle$
5	$\langle x, -1 \rangle$	$\langle 1, -1 \rangle$	$x = \langle 0, 1 \rangle$
6	$\langle x - 1, 0 \rangle$	$\langle -1, -2 \rangle$	$x = \langle 1, 0 \rangle$
7	$\langle x - 1, -1 \rangle$	$\langle 0, -1 \rangle$	$x = \langle 1, 1 \rangle$
8	$\langle x, -y \rangle$	$\langle 4, 0 \rangle$	$y = \langle -x, -x \rangle$
9	$\langle x + y, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x, 0 \rangle$
10	$\langle x - y + 1, -y \rangle$	$\langle 2, 0 \rangle$	$y = \langle 0, -x - 1 \rangle$
11	$\langle x - y + 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x + 1, 0 \rangle$
12	$\langle x + y - 1, y - 1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle 1, x \rangle$
13	$\langle x^2 - y, x^2 \rangle$	$\langle -1, -1 \rangle$	$y = \langle x^2, x^2 \rangle$
14	$\langle x - y, -y + 1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle 1, -x + 1 \rangle$
15	$\langle x + y, y - 1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle 1, x + 1 \rangle$
16	$\langle -y + 1, 1 \rangle$	$\langle -3, -3 \rangle$	$y = \langle 1, 1 \rangle$
17	$\langle -y + 1, 0 \rangle$	$\langle -1, -1 \rangle$	$y = \langle 1, 0 \rangle$
18	$\langle -2x - y, -x - 1 \rangle$	$\langle 4, -4 \rangle$	$y = \langle -2x, -x - 1 \rangle$
19	$\langle -x^2 - y, -x^2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2, -x^2 \rangle$
20	$\langle -x^2 - y + 1, -x^2 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x^2 + 1, -x^2 \rangle$
21	$\langle x^2 - x - y, x^2 - x - 1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle x^2 - x, x^2 - x - 1 \rangle$
22	$\langle x + 1, 0 \rangle$	$\langle -2, -2 \rangle$	$x = \langle -1, 0 \rangle$
23	$\langle x, y \rangle$	$\langle 0, -2 \rangle$	$y = \langle x, x \rangle$
24	$\langle x - y, -y - 1 \rangle$	$\langle 0, -2 \rangle$	$y = \langle -1, -x - 1 \rangle$
25	$\langle -y, x \rangle$	$\langle 0, -1 \rangle$	$y = \langle 0, x \rangle$
26	$\langle x, 1 \rangle$	$\langle -1, -1 \rangle$	$x = \langle 0, -1 \rangle$
27	$\langle -x - y, x^2 - x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x, x^2 - x - 1 \rangle$
28	$\langle 2x - y, x + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 2x, x + 1 \rangle$

Table 2: The polynomials H_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$ of the second certificate. Here $\langle A, B \rangle = A + B\omega$.

i	H_i	c_i	chart
0	$\langle -y, 0 \rangle$	$\langle 4, 0 \rangle$	$y = \langle 0, 0 \rangle$
1	$\langle -y + 1, 1 \rangle$	$\langle -4, -8 \rangle$	$y = \langle 1, 1 \rangle$
2	$\langle -y, -1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle 0, -1 \rangle$
3	$\langle -x - y + 1, x + 1 \rangle$	$\langle -4, -2 \rangle$	$y = \langle -x + 1, x + 1 \rangle$
4	$\langle -2x - y, -x - 1 \rangle$	$\langle 2, -2 \rangle$	$y = \langle -2x, -x - 1 \rangle$
5	$\langle x^2 - y, 0 \rangle$	$\langle 4, 0 \rangle$	$y = \langle x^2, 0 \rangle$
6	$\langle -x^2 - y, -x^2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2, -x^2 \rangle$
7	$\langle x^2 - y, x^2 + x \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2, x^2 + x \rangle$
8	$\langle -x^2 - y + 1, -x^2 - x \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 + 1, -x^2 - x \rangle$
9	$\langle x, 0 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 0, 0 \rangle$
10	$\langle x + 1, 1 \rangle$	$\langle 0, -3 \rangle$	$x = \langle -1, -1 \rangle$
11	$\langle x + 1, 0 \rangle$	$\langle 4, 0 \rangle$	$x = \langle -1, 0 \rangle$
12	$\langle x, -1 \rangle$	$\langle -1, -2 \rangle$	$x = \langle 0, 1 \rangle$
13	$\langle x - 1, 0 \rangle$	$\langle 0, -3 \rangle$	$x = \langle 1, 0 \rangle$
14	$\langle x - 1, -1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 1, 1 \rangle$
15	$\langle x - y, -y \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -x \rangle$
16	$\langle x, -y \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x, -x \rangle$
17	$\langle x + y, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x, 0 \rangle$
18	$\langle x - y + 1, -y \rangle$	$\langle -1, -1 \rangle$	$y = \langle 0, -x - 1 \rangle$

i	H_i	c_i	chart
19	$\langle x + y + 1, y \rangle$	$\langle 0, -1 \rangle$	$y = \langle 0, x + 1 \rangle$
20	$\langle x - y, -y - 1 \rangle$	$\langle -2, -2 \rangle$	$y = \langle -1, -x - 1 \rangle$
21	$\langle x - y - 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x - 1, 0 \rangle$
22	$\langle x + y - 1, y - 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle 1, x \rangle$
23	$\langle -y, x \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, x \rangle$
24	$\langle -y, -x^2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -x^2 \rangle$
25	$\langle x - y + 1, x + 1 \rangle$	$\langle 0, -2 \rangle$	$y = \langle x + 1, x + 1 \rangle$
26	$\langle x^2 + x - y, 0 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x^2 + x, 0 \rangle$
27	$\langle -y + 1, x + 1 \rangle$	$\langle -4, -4 \rangle$	$y = \langle 1, x + 1 \rangle$
28	$\langle x - y + 1, -y + 1 \rangle$	$\langle 4, -4 \rangle$	$y = \langle 1, -x \rangle$
29	$\langle x, 1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 0, -1 \rangle$
30	$\langle -x^2 - y, 0 \rangle$	$\langle 4, 0 \rangle$	$y = \langle -x^2, 0 \rangle$
31	$\langle x - y - 1, -y \rangle$	$\langle 0, -6 \rangle$	$y = \langle 0, -x + 1 \rangle$

Table 3: The data of the identities (2); the entry $j:e$ means $\rho_{ij} = e \neq 0$.

The first certificate.

$i = 0, D_i = 4, \rho_{ij}: 1:2, 4:1, 5:1, 7:1, 10:1, 13:2, 15:1, 16:2, 18:1, 27:2$

$w_i \quad \langle -4t^5 - 36t^4 - 4t^3 + 20t^2 - 12t - 12, \\ 12t^5 + 40t^4 - 12t^2 + 24t + 16 \rangle$

$i = 1, D_i = 1, \rho_{ij}: 0:1, 2:2, 3:2, 5:1, 8:1, 11:1, 12:2, 15:1, 23:1$

$w_i \quad \langle -8t^5 - 7t^4 + 6t^3 + 12t^2 + 5t, \\ -4t^5 - 13t^4 - 11t^3 + 7t + 3 \rangle$

$i = 2, D_i = 18, \rho_{ij}: 1:1, 5:2, 9:2, 16:2, 17:2$

$w_i \quad \langle -18t^4 - 18t^3, \\ -18t^4 - 18t^3 \rangle$

$i = 3, D_i = 2, \rho_{ij}: 1:1, 12:2, 16:1, 18:2, 19:1, 23:1$

$w_i \quad \langle 2t^2 - 4t + 2, \\ 0 \rangle$

$i = 4, D_i = 2, \rho_{ij}: 0:2, 18:2, 23:1, 24:1, 27:2$

$w_i \quad \langle -2t^2 - 4t + 2, \\ -4t - 2 \rangle$

$i = 5, D_i = 4, \rho_{ij}: 0:2, 1:2, 2:1, 9:1, 19:1, 23:2$

$w_i \quad \langle 4t^2 - 4t - 12, \\ -8t^2 - 16t - 16 \rangle$

$i = 6, D_i = 2, \rho_{ij}: 9:2, 10:1, 11:2, 18:1, 21:1, 28:2$

$w_i \quad \langle 2t^2 - 4t, \\ -2t^2 + 4t \rangle$

$i = 7, D_i = 14, \rho_{ij}: 0:2, 9:2, 14:1, 19:1$

$w_i \quad \langle -10t^2 + 20t + 10, \\ -8t^2 + 16t - 6 \rangle$

$i = 8, D_i = 1, \rho_{ij}: 1:2, 9:1, 15:2, 16:1$

$w_i \quad \langle -2t^2 + 2t, \\ 2t^3 - 2t - 2 \rangle$

$i = 9, D_i = 8, \rho_{ij}: 2:1, 5:2, 6:1, 7:1, 8:2, 14:2, 20:2, 22:1$

$w_i \quad \langle -4t^3 - 12t^2 - 16t - 4, \\ 2t^3 - 10t^2 - 16t - 6 \rangle$

$i = 10, D_i = 2, \rho_{ij}: 0:2, 6:2, 20:1, 22:1, 25:2$

$w_i \quad \langle -4t^3 - 4t^2 + 4t + 4, \\ -4t^3 + 8t + 4 \rangle$

$i = 11, D_i = 4, \rho_{ij}: 1:2, 6:1, 15:2, 16:1, 17:2, 22:1, 27:2$

$w_i \quad \langle 15t^3 + 8t^2 - 2t - 1, \\ 17t^3 + 24t^2 + 10t + 1 \rangle$

$i = 12, D_i = 2, \rho_{ij}: 1:1, 3:1, 13:1, 14:1, 28:2$

$w_i \quad \langle -2t + 4, \\ -4t^2 + 2 \rangle$

$i = 13, D_i = 4, \rho_{ij}: 0:1, 12:2, 14:1, 15:2, 16:2, 21:2, 27:2$

$w_i \quad \langle 4t^5 + 4t^3 + 8t^2 - 4, \\ 4t^5 - 4t^3 + 8t^2 + 4t \rangle$

$i = 14, D_i = 2, \rho_{ij}: 7:2, 9:1, 12:2, 13:2, 15:1$

$w_i \quad \langle 4t^2 - 4t, \\ 8t^2 \rangle$

$i = 15, D_i = 2, \rho_{ij}: 0:2, 1:2, 8:1, 11:1, 13:1, 14:2, 19:1$

$w_i \quad \langle -5t^3 - 6t^2 - t + 1, \\$

$-8t^3 - 8t^2 + t + 1\rangle$
 $i = 16, D_i = 2, \rho_{ij}: 0:1, 2:1, 3:2, 8:2, 11:2, 13:1$
 $w_i \quad \langle 6t^3 + 2t^2 - 3t + 1,$
 $\quad \quad -2t^2 - 3t + 2\rangle$
 $i = 17, D_i = 2, \rho_{ij}: 2:1, 11:1, 21:2, 27:1$
 $w_i \quad \langle 2t^3 + 2t^2,$
 $\quad \quad 0\rangle$
 $i = 18, D_i = 2, \rho_{ij}: 0:2, 3:1, 4:1, 6:2, 24:2, 26:1$
 $w_i \quad \langle 8t^3 + 24t^2 - 8t,$
 $\quad \quad -8t^3 + 24t^2 + 8t\rangle$
 $i = 19, D_i = 2, \rho_{ij}: 3:2, 5:2, 7:2, 15:2, 24:1, 26:2$
 $w_i \quad \langle -2t^2 + 2,$
 $\quad \quad -2t^2 + 2t + 2\rangle$
 $i = 20, D_i = 26, \rho_{ij}: 9:1, 10:2, 21:1$
 $w_i \quad \langle -8t^5 + 24t^4 - 30t^3 - 8t^2 + 62t + 32,$
 $\quad \quad 24t^5 + 32t^4 - 40t^3 + 24t^2 + 48t + 8\rangle$
 $i = 21, D_i = 42, \rho_{ij}: 6:2, 13:1, 17:1, 20:2, 25:1$
 $w_i \quad \langle -84t^3 + 42t^2 + 42t,$
 $\quad \quad 0\rangle$
 $i = 22, D_i = 2, \rho_{ij}: 9:2, 10:2, 11:2, 28:2$
 $w_i \quad \langle 4,$
 $\quad \quad 0\rangle$
 $i = 23, D_i = 2, \rho_{ij}: 1:2, 3:2, 4:2, 5:1, 26:1$
 $w_i \quad \langle -2t^3 - 2t^2 + 2,$
 $\quad \quad -2t^2 - 2t\rangle$
 $i = 24, D_i = 6, \rho_{ij}: 4:2, 18:1, 19:2, 26:1, 27:1$
 $w_i \quad \langle 4t^2 + 4,$
 $\quad \quad -4t^3 - 4t^2 - 16t\rangle$
 $i = 25, D_i = 52, \rho_{ij}: 10:1, 21:2$
 $w_i \quad \langle 28t^3 - 30t^2 + 32t + 16,$
 $\quad \quad 60t^3 + 62t^2 + 76t + 38\rangle$
 $i = 26, D_i = 14, \rho_{ij}: 18:2, 19:1, 23:2, 24:2, 28:2$
 $w_i \quad \langle -10t^2 - 8t + 24,$
 $\quad \quad -16t^2 - 10t + 2\rangle$
 $i = 27, D_i = 18, \rho_{ij}: 0:1, 4:1, 11:1, 13:1, 17:2, 24:2$
 $w_i \quad \langle -36t^4 - 36t^3 + 36t^2 + 54t + 18,$
 $\quad \quad -18t^4 - 72t^3 - 54t^2 + 18t + 18\rangle$
 $i = 28, D_i = 2, \rho_{ij}: 6:1, 12:1, 22:1, 26:1$
 $w_i \quad \langle 8t^2 - 2t - 2,$
 $\quad \quad -4t^3 + 6t^2 + 4t - 2\rangle$
The second certificate.
 $i = 0, D_i = 1, \rho_{ij}: 15:1, 26:1$
 $w_i \quad \langle 2t^2,$
 $\quad \quad 2t^2\rangle$
 $i = 1, D_i = 1, \rho_{ij}: 6:2, 11:2, 12:1, 22:2$
 $w_i \quad \langle 2t + 1,$
 $\quad \quad t - 1\rangle$
 $i = 2, D_i = 4, \rho_{ij}: 5:2, 6:1, 7:2, 8:1, 10:1, 11:1, 13:1, 17:1, 18:1, 19:2, 21:1, 26:1, 29:2$
 $w_i \quad \langle -26t^3 - 70t^2 - 36t + 36,$
 $\quad \quad 44t^3 - 26t^2 - 60t + 12\rangle$
 $i = 3, D_i = 2, \rho_{ij}: 4:2, 8:1, 16:1, 21:2, 23:1, 25:2$
 $w_i \quad \langle 16t^3 + 16t^2,$
 $\quad \quad 20t^3 + 8t^2 - 12t\rangle$
 $i = 4, D_i = 2, \rho_{ij}: 3:1, 5:1, 17:1, 20:1, 21:1, 22:1, 23:1, 29:2$
 $w_i \quad \langle -8t^3 - 32t^2 + 8t + 8,$
 $\quad \quad 8t^3 - 28t^2 - 20t + 4\rangle$
 $i = 5, D_i = 2, \rho_{ij}: 2:1, 4:2, 10:1, 14:2, 16:1, 18:1, 19:1, 23:2, 24:1, 25:1, 31:1$
 $w_i \quad \langle -8t^5 - 16t^4 + 8t^3 + 4t^2 - 12t - 4,$
 $\quad \quad -4t^5 - 4t^4 - 4t^2 - 4t\rangle$
 $i = 6, D_i = 4, \rho_{ij}: 1:1, 2:2, 12:1, 17:2, 19:1, 20:1, 27:2, 30:2$
 $w_i \quad \langle -4t^5 + 4t^4 - 4t^3 - 4t^2 + 4,$
 $\quad \quad -4t^5 - 4t^4 - 8t^3 - 4t^2 - 4t\rangle$
 $i = 7, D_i = 2, \rho_{ij}: 2:1, 8:1, 9:2, 14:1, 16:2, 17:1, 19:1, 22:2, 25:1, 29:1$
 $w_i \quad \langle -4t^5 + 4t^4 - 2t^2 + 4t + 2,$
 $\quad \quad -8t^5 + 2t^3 - 6t^2 + 2\rangle$
 $i = 8, D_i = 12500, \rho_{ij}: 2:2, 3:2, 7:2, 9:1, 13:1, 15:2, 17:2, 19:2, 21:2, 22:2, 29:1$
 $w_i \quad \langle 425000t^5 + 2012500t^4 - 1612500t^3 - 2175000t^2 + 300000t + 300000,$
 $\quad \quad -112500t^5 + 1887500t^4 + 1137500t^3 - 1537500t^2 - 900000t - 75000\rangle$

$i = 9, D_i = 2, \rho_{ij}: 7:1, 8:2, 21:1$
$w_i \quad \langle -2t, \quad 0 \rangle$
$i = 10, D_i = 2, \rho_{ij}: 2:2, 5:2, 15:2, 27:1, 30:2$
$w_i \quad \langle 2t^2 + 4, \quad -2t^2 + 2 \rangle$
$i = 11, D_i = 1, \rho_{ij}: 1:1, 2:2, 18:1, 20:2, 21:2, 24:2, 27:1$
$w_i \quad \langle -2t^2 - 2t, \quad -2t - 2 \rangle$
$i = 12, D_i = 4, \rho_{ij}: 1:2, 6:2, 23:1, 25:2, 26:2, 30:2$
$w_i \quad \langle 12t^2 + 24t + 28, \quad 12t^2 + 24t + 8 \rangle$
$i = 13, D_i = 4, \rho_{ij}: 2:2, 8:2, 18:2, 21:2, 22:2, 25:1, 26:2$
$w_i \quad \langle 48t^2 - 96t + 144, \quad 24t^2 - 48t + 48 \rangle$
$i = 14, D_i = 2, \rho_{ij}: 5:1, 7:2, 15:2, 21:2, 22:1, 26:1$
$w_i \quad \langle 4t + 2, \quad 2t^2 + 4t - 4 \rangle$
$i = 15, D_i = 2, \rho_{ij}: 0:2, 8:1, 10:1, 14:1, 22:2, 27:1$
$w_i \quad \langle -4t^3 + 2t^2, \quad 2t^2 \rangle$
$i = 16, D_i = 4, \rho_{ij}: 3:2, 5:2, 7:1, 21:2, 25:1, 26:1, 27:2, 28:2, 29:2$
$w_i \quad \langle -29t^3 - 51t^2 + 2t + 1, \quad 12t^3 - 31t^2 - 33t - 18 \rangle$
$i = 17, D_i = 160, \rho_{ij}: 2:2, 4:2, 6:1, 7:2, 8:1, 19:2, 20:2, 21:1, 22:2, 30:2$
$w_i \quad \langle 722t^3 + 78t^2 - 960t - 398, \quad 1500t^3 + 420t^2 - 1280t - 580 \rangle$
$i = 18, D_i = 4, \rho_{ij}: 2:2, 5:2, 11:2, 13:1, 19:2, 23:2, 24:1, 26:1, 30:2$
$w_i \quad \langle -39t^3 - 44t^2 - 12t + 11, \quad 17t^3 + 7t^2 + 22t + 22 \rangle$
$i = 19, D_i = 8, \rho_{ij}: 2:1, 5:2, 6:2, 7:2, 8:1, 17:1, 18:1, 21:2$
$w_i \quad \langle -4t^3 + 57t^2 + 86t + 45, \quad 5t^3 + 40t^2 + 22t + 1 \rangle$
$i = 20, D_i = 6, \rho_{ij}: 4:2, 6:2, 11:1, 17:1, 21:2$
$w_i \quad \langle -8t^3 + 8t^2 + 28t + 16, \quad 16t^2 + 12t - 8 \rangle$
$i = 21, D_i = 8, \rho_{ij}: 2:2, 3:1, 4:2, 8:1, 9:2, 11:1, 13:1, 14:1, 16:1, 17:2, 19:1, 20:1, 26:2, 31:1$
$w_i \quad \langle 45t^3 - 42t^2 - 32t + 41, \quad -520t^3 + 552t^2 - 288t - 160 \rangle$
$i = 22, D_i = 2, \rho_{ij}: 1:1, 4:2, 7:1, 8:1, 13:1, 14:2, 15:1, 17:1, 28:2$
$w_i \quad \langle 12t^3 - 20t^2 + 8t, \quad -4t^3 - 8t^2 + 4t - 4 \rangle$
$i = 23, D_i = 52, \rho_{ij}: 3:2, 4:2, 5:1, 12:2, 18:1, 26:2, 28:2$
$w_i \quad \langle -316t^3 - 148t^2 + 196t + 124, \quad -120t^3 + 188t^2 + 160t + 80 \rangle$
$i = 24, D_i = 4, \rho_{ij}: 5:2, 11:1, 18:2, 28:1, 31:2$
$w_i \quad \langle 4t^3 - 4t^2, \quad -4t^5 + 4t^4 + 8t^3 - 8t^2 + 4 \rangle$
$i = 25, D_i = 4, \rho_{ij}: 3:1, 5:2, 7:2, 12:1, 13:2, 16:2, 30:2$
$w_i \quad \langle 10t^3 + 2t^2 + 4t + 8, \quad -18t^3 - 12t^2 + 6 \rangle$
$i = 26, D_i = 324, \rho_{ij}: 0:2, 2:2, 12:1, 13:1, 14:2, 16:2, 18:2, 21:1, 23:1, 28:1, 30:1, 31:2$
$w_i \quad \langle -6804t^5 - 5832t^4 - 1620t^3 + 648t^2 - 1296t - 648, \quad -5508t^5 - 22032t^4 - 11340t^3 - 648t^2 + 2268t - 5508 \rangle$
$i = 27, D_i = 1, \rho_{ij}: 6:1, 10:2, 11:2, 15:2, 16:1, 28:1$
$w_i \quad \langle 4t^3 + 6t^2 + 4t + 2, \quad 4t^3 + 8t^2 + 6t + 2 \rangle$
$i = 28, D_i = 2, \rho_{ij}: 16:1, 22:1, 23:1, 24:2, 26:2, 27:2$
$w_i \quad \langle 16t^3 + 20t^2 + 4t, \quad 8t^3 + 16t^2 + 8t \rangle$
$i = 29, D_i = 2, \rho_{ij}: 2:1, 4:1, 7:2, 8:2, 16:1$
$w_i \quad \langle 2, \quad -2t + 2 \rangle$
$i = 30, D_i = 10, \rho_{ij}: 6:1, 10:1, 12:1, 17:1, 18:1, 25:1, 26:2$
$w_i \quad \langle 8t^5 + 24t^4 - 20t^3 + 32t^2 + 40t + 24, \quad 4t^5 + 32t^4 + 16t^2 + 20t + 12 \rangle$
$i = 31, D_i = 2, \rho_{ij}: 5:2, 21:2, 24:1, 26:1$
$w_i \quad \langle -2t^3 + 4t - 2, \quad \quad \quad \rangle$

Table 4: The edges (i, j) with a linear endpoint, the linear endpoint k used, and the norm of $R_{ij} = \text{Res}_t(f_k, g_{kl})$.

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
<i>The first certificate.</i>					
(0, 1)	0	2^4	(7, 9)	7	7
(0, 4)	0	2^4	(7, 14)	7	$2^2 \cdot 7$
(0, 5)	0	2^4	(7, 19)	7	1
(0, 7)	0	2^2	(8, 9)	8	1
(0, 10)	0	2^4	(8, 15)	8	2^4
(0, 13)	0	1	(8, 16)	8	3
(0, 15)	0	2^2	(9, 14)	9	7^3
(0, 16)	0	$2^2 \cdot 3^2$	(9, 20)	9	3
(0, 18)	0	$2^8 \cdot 3^2$	(9, 22)	9	1
(0, 27)	0	1	(10, 20)	10	7
(1, 2)	1	3	(10, 22)	10	2^2
(1, 3)	1	1	(10, 25)	10	$2^4 \cdot 7$
(1, 5)	1	$2^2 \cdot 7$	(11, 15)	11	2^4
(1, 8)	1	$2^8 \cdot 3$	(11, 16)	11	$2^2 \cdot 3$
(1, 11)	1	2^4	(11, 17)	11	1
(1, 12)	1	7	(11, 22)	11	2^2
(1, 15)	1	2^2	(11, 27)	11	1
(1, 23)	1	2^4	(12, 13)	12	$3^3 \cdot 7$
(2, 5)	2	1	(12, 14)	12	$2^6 \cdot 7$
(2, 9)	2	1	(12, 28)	12	2^2
(2, 16)	2	3	(13, 14)	13	$2^4 \cdot 7$
(2, 17)	2	1	(13, 15)	13	2^4
(3, 12)	3	1	(13, 16)	13	3^3
(3, 16)	3	1	(13, 21)	13	3
(3, 18)	3	1	(13, 27)	13	3^2
(3, 19)	3	1	(14, 15)	14	2^8
(3, 23)	3	1	(15, 19)	15	2^4
(4, 18)	4	$3 \cdot 7$	(17, 21)	17	1
(4, 23)	4	$2^2 \cdot 3$	(17, 27)	17	1
(4, 24)	4	$2^2 \cdot 7$	(18, 24)	18	$2^6 \cdot 3^2 \cdot 7$
(4, 27)	4	1	(18, 26)	18	3
(5, 9)	5	$3 \cdot 7$	(19, 24)	19	$2^4 \cdot 7$
(5, 19)	5	1	(19, 26)	19	7
(5, 23)	5	3	(20, 21)	20	$3^5 \cdot 7$
(6, 9)	6	3^2	(21, 25)	21	7
(6, 10)	6	2^4	(22, 28)	22	2^2
(6, 11)	6	$2^4 \cdot 3^2$	(23, 26)	23	1
(6, 18)	6	$2^4 \cdot 3^2$	(24, 26)	24	$3^2 \cdot 7$
(6, 21)	6	3^2	(24, 27)	24	$3^2 \cdot 7^3$
(6, 28)	6	2^4	(26, 28)	26	1
<i>The second certificate.</i>					
(0, 15)	0	1	(8, 17)	8	7
(0, 26)	0	2^2	(8, 19)	8	$2^4 \cdot 7^2$
(1, 6)	1	3^2	(8, 21)	8	$2^2 \cdot 7 \cdot 13$
(1, 11)	1	3	(8, 22)	8	$5^2 \cdot 7^2$
(1, 12)	1	$2^2 \cdot 3$	(8, 29)	8	1
(1, 22)	1	3^2	(9, 21)	9	1
(2, 5)	2	$2^2 \cdot 3^2$	(10, 15)	10	3
(2, 6)	2	$3^2 \cdot 13$	(10, 27)	10	2^2

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(2, 7)	2	$7 \cdot 13$	(10, 30)	10	$2^2 \cdot 3$
(2, 8)	2	7^2	(11, 18)	11	2^2
(2, 10)	2	$2^2 \cdot 3$	(11, 20)	11	3^2
(2, 11)	2	3	(11, 21)	11	2^2
(2, 13)	2	3^2	(11, 24)	11	3
(2, 17)	2	$3 \cdot 7$	(11, 27)	11	1
(2, 18)	2	1	(12, 23)	12	2^4
(2, 19)	2	$3^2 \cdot 7 \cdot 13$	(12, 25)	12	1
(2, 21)	2	$2^2 \cdot 7 \cdot 13$	(12, 26)	12	3
(2, 26)	2	$3 \cdot 7$	(12, 30)	12	$2^2 \cdot 3$
(2, 29)	2	3	(13, 18)	13	2^4
(3, 4)	3	$2^2 \cdot 3^6$	(13, 21)	13	2^4
(3, 8)	3	$2^4 \cdot 7 \cdot 13$	(13, 22)	13	3^2
(3, 16)	3	$2^2 \cdot 3 \cdot 13$	(13, 25)	13	2^4
(3, 21)	3	$2^2 \cdot 7 \cdot 13^2$	(13, 26)	13	$2^4 \cdot 3^2$
(3, 23)	3	3	(14, 15)	14	1
(3, 25)	3	2^8	(14, 21)	14	2^2
(4, 5)	4	$2^8 \cdot 3^2$	(14, 22)	14	1
(4, 17)	4	$3 \cdot 7$	(14, 26)	14	7^2
(4, 20)	4	$2^6 \cdot 3^2 \cdot 7$	(15, 22)	15	$2^2 \cdot 3 \cdot 5^2$
(4, 21)	4	$2^2 \cdot 7 \cdot 13^2$	(15, 27)	15	2^2
(4, 22)	4	$2^4 \cdot 3^3$	(16, 21)	16	$2^4 \cdot 13$
(4, 23)	4	$2^2 \cdot 3 \cdot 13$	(16, 25)	16	$2^4 \cdot 7$
(4, 29)	4	3	(16, 26)	16	$7 \cdot 13$
(5, 10)	5	2^4	(16, 27)	16	2^4
(5, 14)	5	2^2	(16, 28)	16	3
(5, 16)	5	2^4	(16, 29)	16	3
(5, 18)	5	2^4	(17, 19)	17	$3 \cdot 7$
(5, 19)	5	2^4	(17, 20)	17	7^3
(5, 23)	5	2^4	(17, 21)	17	$2^4 \cdot 7^2$
(5, 24)	5	1	(17, 22)	17	$3 \cdot 7$
(5, 25)	5	2^4	(17, 30)	17	3^2
(5, 31)	5	2^4	(18, 19)	18	2^{10}
(6, 12)	6	1	(18, 23)	18	$2^4 \cdot 7$
(6, 17)	6	7	(18, 24)	18	1
(6, 19)	6	13	(18, 26)	18	$2^2 \cdot 7^2$
(6, 20)	6	$2^4 \cdot 7$	(18, 30)	18	$2^4 \cdot 7$
(6, 27)	6	2^4	(19, 21)	19	$2^2 \cdot 7 \cdot 13$
(6, 30)	6	1	(20, 21)	20	$2^2 \cdot 7$
(7, 8)	7	$5^2 \cdot 13$	(21, 26)	21	$2^2 \cdot 13^2$
(7, 9)	7	1	(21, 31)	21	2^4
(7, 14)	7	7^2	(22, 28)	22	2^8
(7, 16)	7	$7 \cdot 13$	(23, 26)	23	$7 \cdot 13$
(7, 17)	7	1	(23, 28)	23	$2^2 \cdot 3 \cdot 13$
(7, 19)	7	13^2	(24, 28)	24	3^4
(7, 22)	7	5^2	(24, 31)	24	3^2
(7, 25)	7	7	(25, 30)	25	$2^4 \cdot 7$
(7, 29)	7	1	(26, 28)	26	$3^2 \cdot 13$
(8, 9)	8	1	(26, 30)	26	$3^4 \cdot 7^2$
(8, 13)	8	2^4	(26, 31)	26	$2^2 \cdot 3^2$
(8, 15)	8	5^2	(27, 28)	27	2^6

Table 5: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of $(B_p^{(g)})_g$.

p	k	value	classes (a, b)
2	3	(1, 2)	(5, 0), (5, 2), (5, 4), (5, 6)
3	3	(0, 1)	(22, 2), (22, 11), (22, 20)
3	3	(0, 2)	(5, 17), (14, 8), (23, 26)
3	3	(1, 1)	(2, 11), (11, 2), (20, 20)
3	3	(1, 2)	(4, 8), (4, 17), (4, 26)
3	3	(2, 0)	(8, 14), (13, 5), (13, 14), (13, 23), (17, 5), (26, 23)
5	1	(0, 0)	(0, 3), (1, 1), (2, 0), (2, 2), (2, 3)
7	1	(0, 0)	(0, 2)
7	2	(0, 0)	(2, 14), (3, 11), (5, 45), (9, 0), (10, 32), (16, 35), (17, 4), (18, 10), (19, 31), (23, 21), (24, 25), (25, 17), (26, 24), (30, 7), (31, 19), (31, 33), (31, 46), (32, 24), (33, 17), (37, 42), (38, 18), (39, 31), (40, 10), (44, 28), (45, 39), (46, 38), (47, 3)
7	3	(0, 0)	(4, 290), (11, 248), (12, 136), (53, 339), (60, 297), (61, 87), (102, 45), (109, 3), (110, 38), (129, 26), (129, 75), (129, 124), (129, 173), (129, 222), (129, 271), (129, 320), (151, 94), (158, 52), (159, 332), (178, 5), (178, 47), (178, 54), (178, 96), (178, 103), (178, 145), (178, 152), (178, 194), (178, 201), (178, 243), (178, 250), (178, 292), (178, 299), (178, 341), (200, 143), (207, 101), (208, 283), (227, 12), (227, 40), (227, 61), (227, 89), (227, 110), (227, 138), (227, 159), (227, 187), (227, 208), (227, 236), (227, 257), (227, 285), (227, 306), (227, 334), (249, 192), (256, 150), (257, 234), (298, 241), (305, 199), (306, 185)
13	1	(0, 0)	(2, 3), (3, 1), (4, 8), (4, 10), (5, 1), (6, 7), (7, 0), (9, 2), (10, 4), (12, 5)
13	2	(0, 0)	(8, 163), (11, 166), (21, 137), (24, 75), (34, 111), (37, 153), (47, 85), (50, 62), (63, 140), (73, 33), (76, 49), (86, 7), (99, 150), (102, 36), (112, 124), (115, 114), (125, 98), (128, 23), (138, 72), (141, 101), (151, 46), (154, 10), (164, 20), (167, 88)
13	3	(0, 0)	(60, 59), (89, 972), (229, 1918), (258, 1986), (398, 1580), (427, 803), (567, 1242), (596, 1817), (736, 904), (765, 634), (905, 566), (934, 1648), (1074, 228), (1103, 465), (1243, 2087), (1272, 1479), (1412, 1749), (1441, 296), (1581, 1411), (1610, 1310), (1750, 1073), (1779, 127), (1919, 735), (1948, 1141), (2088, 397), (2117, 2155)